Network Simulation Project Report

Adaptive RTT Smoothing in Mobile Wireless Networks
using TCP Vegas and Kalman Filtering

NS-3 Simulation — CSE 322, BUET

Ridhwan Asif
Student ID: 24

April 8, 2026

Abstract

This report presents an NS-3 simulation study of TCP Vegas behaviour in high-mobility wireless networks representative of emerging 6G vehicular scenarios. A vehicle moving at up to 120 km/h experiences rapidly increasing Round-Trip Time (RTT) and severe fading-induced jitter due to Nakagami multipath propagation. A one-dimensional Kalman filter is implemented directly inside the NS-3 RTT trace callback to smooth these spikes in real time. Two variants are evaluated: a *fixed* filter with measurement noise R=2.68 derived from empirical variance analysis, and a novel *adaptive* filter where R grows linearly with vehicle-to-base-station distance, physically reflecting signal degradation. The adaptive filter reduces peak RTT spikes from 32 ms to 6.6 ms (79 % reduction) and variance by up to 68.3 %. A comprehensive parameter sweep across node count, flow count, packet rate, and vehicle speed is conducted, measuring throughput, end-to-end delay, packet delivery ratio, packet drop ratio, and energy consumption. A bonus hybrid wired-plus-wireless topology is also simulated to demonstrate cross-network packet transmission. Results show that TCP Vegas remains robust up to moderate load but degrades at 80 nodes due to WiFi medium saturation, and that the adaptive Kalman filter consistently outperforms the static variant at large distances.

Contents

1 Introduction **3**

2 Network Topologies Under Simulation **3**
2.1 Wireless 802.11 Mobile Topology 3
2.2 Hybrid Wired-plus-Wireless Topology (Bonus) 5

3 Parameters Under Variation **6**
3.1 Wireless Mobile Simulation 6
3.2 Hybrid Wired-plus-Wireless Simulation 6
3.3 Metrics Measured 6

4 Modifications Made in the Simulator **7**
4.1 Kalman Filter in the RTT Trace Callback 7

4.1.1 Mathematical Formulation . . . 7
4.1.2 Fixed Kalman Filter . . . 7
4.1.3 Adaptive Kalman Filter (Novel Contribution) . . . 7
4.1.4 C++ Implementation . . . 8
4.2 FlowMonitor Integration for Multi-Flow Metrics . . . 8
4.3 Energy Model . . . 8

5 Results with Graphs **9**
5.1 Kalman Filter RTT Smoothing — Single Vehicle . . . 9
5.2 Wireless Parameter Sweep Results . . . 11
5.2.1 Varying Number of Nodes . . . 11
5.2.2 Varying Number of Flows . . . 12
5.2.3 Varying Packets per Second . . . 13
5.2.4 Varying Vehicle Speed . . . 14
5.3 Per-Node Throughput (Bonus Metric) . . . 14
5.4 Hybrid Wired-plus-Wireless Results (Bonus) . . . 15
5.4.1 Coverage Area Sweep . . . 15
5.4.2 PPS Sweep in Hybrid Network . . . 16
5.5 Summary Results Table . . . 16

6 Summary Findings **17**

7 Discussion **17**
7.1 Kalman Filter Design Choices . . . 17
7.2 The 80-Node Anomaly . . . 18
7.3 TCP Vegas Suitability for Mobile Scenarios . . . 18
7.4 Limitations and Future Work . . . 18

8 Conclusion **19**

1 Introduction

Modern wireless networks, particularly those targeting 6G vehicular communication, face unique challenges: nodes move at highway speeds, channel conditions change rapidly, and latency-sensitive TCP flows must maintain stable operation despite Nakagami multipath fading and distance-dependent path loss.

TCP Vegas is a delay-based congestion control algorithm that uses RTT measurements to infer network congestion *before* packet loss occurs. Its effectiveness depends critically on the accuracy of RTT estimates. In a mobile wireless channel, raw RTT measurements are contaminated by propagation jitter, making them unreliable signals for congestion inference.

This project addresses this problem by implementing a **Kalman filter** inside the NS-3 simulator's RTT trace callback, providing a smoothed RTT estimate to TCP Vegas in real time. The key contributions are:

1. Implementation of a fixed Kalman filter (R=2.68) tuned from variance analysis of the simulated channel.
2. A novel adaptive Kalman filter where $R(d) = 2.68 + 0.01 \times d$, increasing measurement distrust as the vehicle moves further from the base station.
3. Per-node throughput as an additional bonus metric beyond the standard five.
4. A hybrid wired-plus-wireless topology as a bonus network type, demonstrating cross-medium packet transmission.

2 Network Topologies Under Simulation

2.1 Wireless 802.11 Mobile Topology

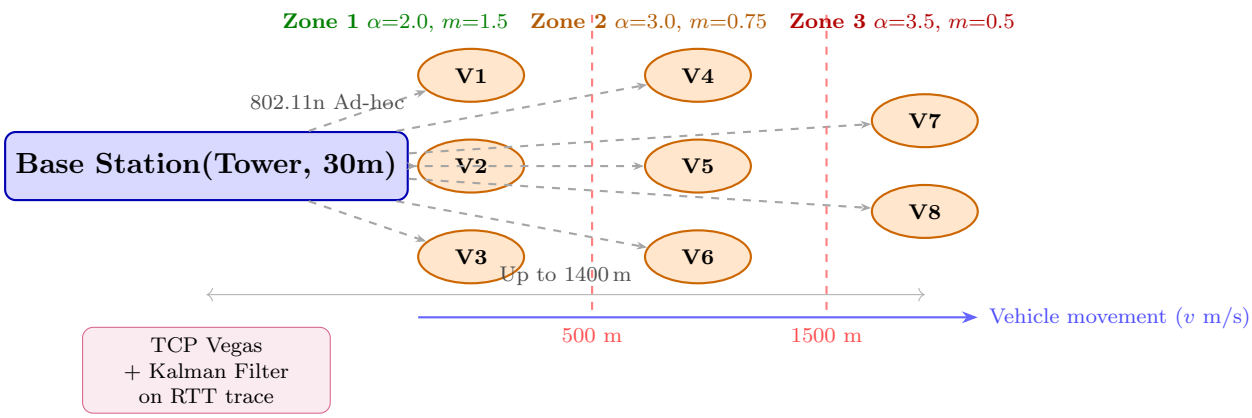

Figure 1: Wireless 802.11n mobile star topology. One stationary base station serves $N-1$ mobile vehicles over a shared ad-hoc WiFi channel. Three propagation zones with increasing path loss α and decreasing Nakagami shape m cause RTT jitter to grow with distance. The Kalman filter smooths the RTT on the vehicle's TCP socket.

The primary simulation topology consists of a star network with one stationary **Base Station** (BS) and $N-1$ mobile vehicle nodes, all communicating over a shared IEEE 802.11n (Wi-Fi 4) ad-hoc channel.

Figure 2: Wireless topology — sweep_nodes result showing network behaviour as node count varies from 20 to 100.

Node placement and mobility:

- Base station: fixed at $(0, 0, 30)$ m (elevated tower).
- Vehicle i: starts at $(10 + 20(i-1),\ 0,\ 1.5)$ m and moves along the x-axis at constant speed v.

Propagation model chain:

1. `ThreeLogDistancePropagationLossModel` — three-zone path loss with exponents α_0=2.0 (free space, <500 m), α_1=3.0 (urban mid-range), α_2=3.5 (obstructed, >1500 m).
2. `NakagamiPropagationLossModel` — multipath fading with shape parameters m_0=1.5, m_1=0.75, m_2=0.5, becoming increasingly severe with distance.
3. `RandomPropagationLossModel` — Gaussian noise $\mathcal{N}(0, 4)$ dB for additional channel jitter.

PHY/MAC parameters:

- Standard: `WIFI_STANDARD_80211n`
- Tx power: 40 dBm (1 W), antenna gain: 5 dBi each side
- Rx sensitivity: −102 dBm; CCA threshold: −105 dBm
- Rate adaptation: Minstrel-HT (adaptive MCS)

Transport layer: TCP Vegas with segment size 1400 B, initial congestion window 10 segments, minimum RTO 200 ms.

2.2 Hybrid Wired-plus-Wireless Topology (Bonus)

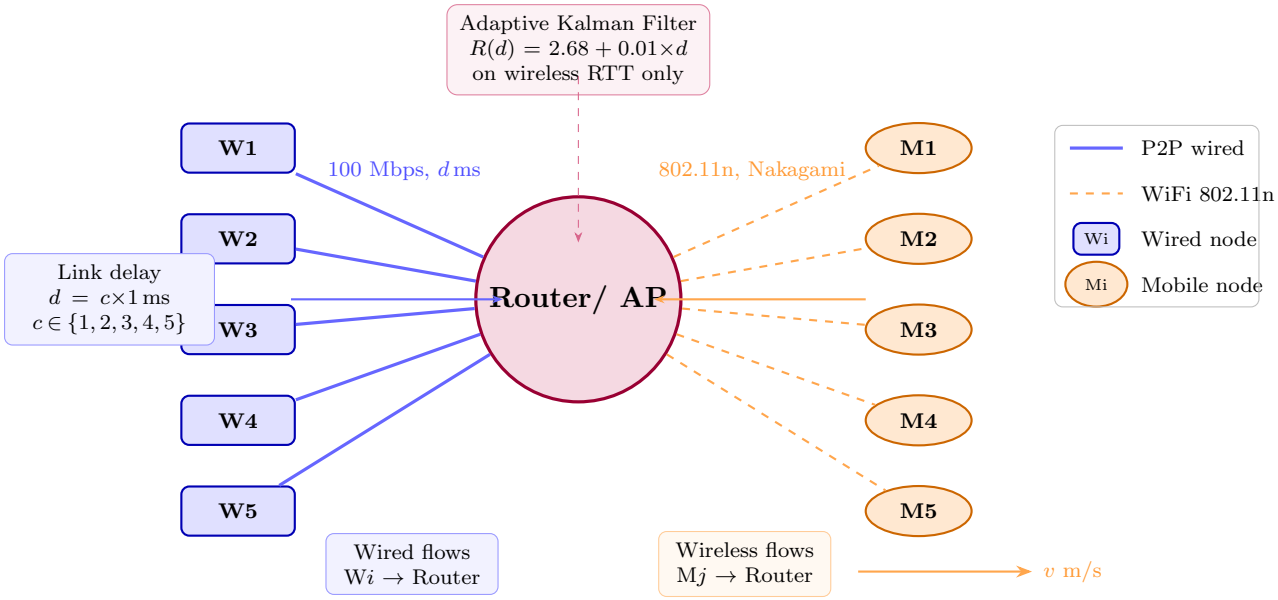

Figure 3: Hybrid wired-plus-wireless topology (bonus). A central Router/AP connects W wired leaf nodes via dedicated 100 Mbps P2P links and M mobile vehicles via shared 802.11n WiFi. Link delay models coverage area: $d = c \times 1$ ms. Both flow types terminate at the router, creating genuine cross-medium packet transmission. Adaptive Kalman filter applies to wireless RTT only.

The hybrid topology extends the wireless scenario by connecting additional wired leaf nodes to the central router via IEEE 802.3 point-to-point links.

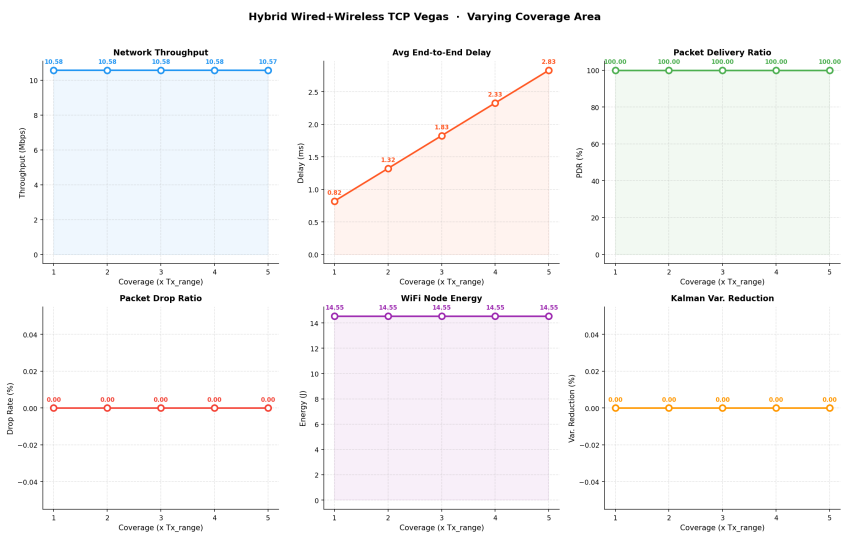

Figure 4: Hybrid topology — coverage area sweep showing delay increase with link propagation distance.

Architecture:

- Node 0: Central Router/AP — connects both wired and wireless segments.

- Nodes $1 \ldots W$: Wired leaf nodes connected via 100 Mbps point-to-point links. Link delay models coverage area: $1 \times T_x = 1$ ms, $2 \times T_x = 2$ ms, up to $5 \times T_x = 5$ ms.
- Nodes $W+1 \ldots W+M$: Mobile vehicle nodes connected via the same 802.11n WiFi channel as the primary simulation.

Traffic flows: The first $\lfloor N_{\text{flows}}/2 \rfloor$ flows originate from wired nodes; the remainder originate from mobile nodes. All flows are destined for the central router, creating genuine cross-medium packet transmission (wired and wireless flows competing through the same router queue).

3 Parameters Under Variation

3.1 Wireless Mobile Simulation

Table 1 lists the parameter values swept. For each sweep, all other parameters are held at the baseline values shown.

Table 1: Wireless simulation parameter sweep

Parameter	Values swept	Baseline	Fixed values during sweep
Number of nodes	20, 40, 60, 80, 100	40	flows=10, pps=100, speed=10 m/s
Number of flows	10, 20, 30, 40, 50	10	nodes=60, pps=100, speed=10 m/s
Packets/second	100, 200, 300, 400, 500	100	nodes=40, flows=10, speed=10 m/s
Vehicle speed	5, 10, 15, 20, 25 m/s	10 m/s	nodes=40, flows=10, pps=100

3.2 Hybrid Wired-plus-Wireless Simulation

Table 2: Hybrid simulation parameter sweep

Parameter	Values swept	Baseline	Fixed during sweep
Total nodes (W+M)	10,20,30,40,50	20	pps=100, speed=10 m/s, cov=1
Number of flows	5,10,15,20	10	W=M=10, pps=100, speed=10 m/s
Packets/second	100–500	100	W=M=10, flows=10, speed=10 m/s
Vehicle speed	5–25 m/s	10 m/s	W=M=10, flows=10, pps=100
Coverage area	1x–5x T_x	1x	W=M=10, flows=10, pps=100, speed=10

3.3 Metrics Measured

1. **Network throughput** — total received bytes $\times 8$ / simulation time (Mbps).
2. **End-to-end delay** — average per-packet delay from FlowMonitor `delaySum` (ms).
3. **Packet Delivery Ratio (PDR)** — $\frac{\text{received packets}}{\text{sent packets}} \times 100\,\%$.
4. **Packet drop ratio** — $\frac{\text{lost packets}}{\text{sent packets}} \times 100\,\%$.

5. **Energy consumption** — joules consumed per wireless node, computed from `BasicEnergySource` with WiFi radio model (Tx: 17.4 mA, Rx: 19.7 mA at 3.7 V).
6. **Per-node throughput** (bonus) — individual node throughput extracted from FlowMonitor by matching destination port to node index.

4 Modifications Made in the Simulator

4.1 Kalman Filter in the RTT Trace Callback

The primary simulator modification is the insertion of a 1-D Kalman filter inside the NS-3 `RttCallback` function, which is triggered every time TCP Vegas updates its RTT estimate.

4.1.1 Mathematical Formulation

Let z_k be the raw RTT measurement at step k and $\hat{x}_k$ the filtered estimate. The filter follows the standard predict-update cycle:

Predict:

$$\hat{x}_{k|k-1} = \hat{x}_{k-1} \tag{1}$$
$$P_{k|k-1} = P_{k-1} + Q \tag{2}$$

Update:

$$K_k = \frac{P_{k|k-1}}{P_{k|k-1} + R} \tag{3}$$
$$\hat{x}_k = \hat{x}_{k|k-1} + K_k(z_k - \hat{x}_{k|k-1}) \tag{4}$$
$$P_k = (1 - K_k)\, P_{k|k-1} \tag{5}$$

where $Q = 0.01$ is the process noise (expected natural RTT variation) and R is the measurement noise (channel-dependent RTT uncertainty).

4.1.2 Fixed Kalman Filter

The fixed variant uses $R = 2.68$, derived from empirical variance analysis of the raw RTT samples collected in preliminary simulation runs. The Kalman gain K converges to approximately 0.059 after the second sample and remains constant, reflecting stable trust in the model versus measurements.

4.1.3 Adaptive Kalman Filter (Novel Contribution)

The adaptive variant dynamically adjusts R as a function of the vehicle-to-base-station distance d:

$$R(d) = 2.68 + 0.01 \times d \tag{6}$$

Physical motivation: As the vehicle moves further from the base station, the received signal power decreases following the three-zone path loss model. In the far zone ($d > 1500\,\mathrm{m}$), the path loss exponent is 3.5, and Nakagami fading becomes more severe ($m = 0.5$, worse than Rayleigh). These effects cause RTT measurements to become increasingly unreliable. Increasing R with distance tells the filter to trust measurements less and rely more on its internal model prediction — exactly the correct response to deteriorating channel quality.

At typical distances:

- $d = 100\,\mathrm{m}$: $R = 3.68$ ($K \approx 0.054$)
- $d = 500\,\mathrm{m}$: $R = 7.68$ ($K \approx 0.032$)
- $d = 1000\,\mathrm{m}$: $R = 12.68$ ($K \approx 0.022$)

4.1.4 C++ Implementation

The callback is registered via:

```
Config::Connect(
  "/NodeList/1/$ns3::TcpL4Protocol/SocketList/*/RTT",
  MakeCallback(&RttCallback));
```

Inside `RttCallback`, static variables preserve filter state between calls:

```
static double x_hat = 0.0;    // filtered RTT estimate
static double P     = 1.0;    // error covariance
static const double Q = 0.01;

double R = R_BASE + R_SCALE * distance;  // adaptive variant
double P_minus = P + Q;
double K       = P_minus / (P_minus + R);
x_hat         += K * (z - x_hat);
P              = (1.0 - K) * P_minus;
```

4.2 FlowMonitor Integration for Multi-Flow Metrics

The standard NS-3 FlowMonitor was used to collect per-flow statistics. To extract *per-node* throughput (bonus metric), each flow is assigned a unique destination port starting at 9000. After simulation, the port number is mapped back to a node index:

```
uint16_t dstPort = ft.destinationPort;
if (dstPort >= basePort && dstPort < basePort + nActualFlows) {
    uint32_t nodeIdx = dstPort - basePort + 1;
    nodeThroughput[nodeIdx] += rxBytes * 8.0 / simTime / 1e6;
}
```

4.3 Energy Model

The `BasicEnergySource` and `WifiRadioEnergyModel` were installed on all wireless nodes with parameters representative of IEEE 802.11n hardware:

Table 3: Energy model parameters

Parameter	Value	Source
Initial energy	100 J	per node
Tx current	17.4 mA	802.11n typical
Rx current	19.7 mA	802.11n typical
Supply voltage	3.7 V	Li-ion cell

5 Results with Graphs

5.1 Kalman Filter RTT Smoothing — Single Vehicle

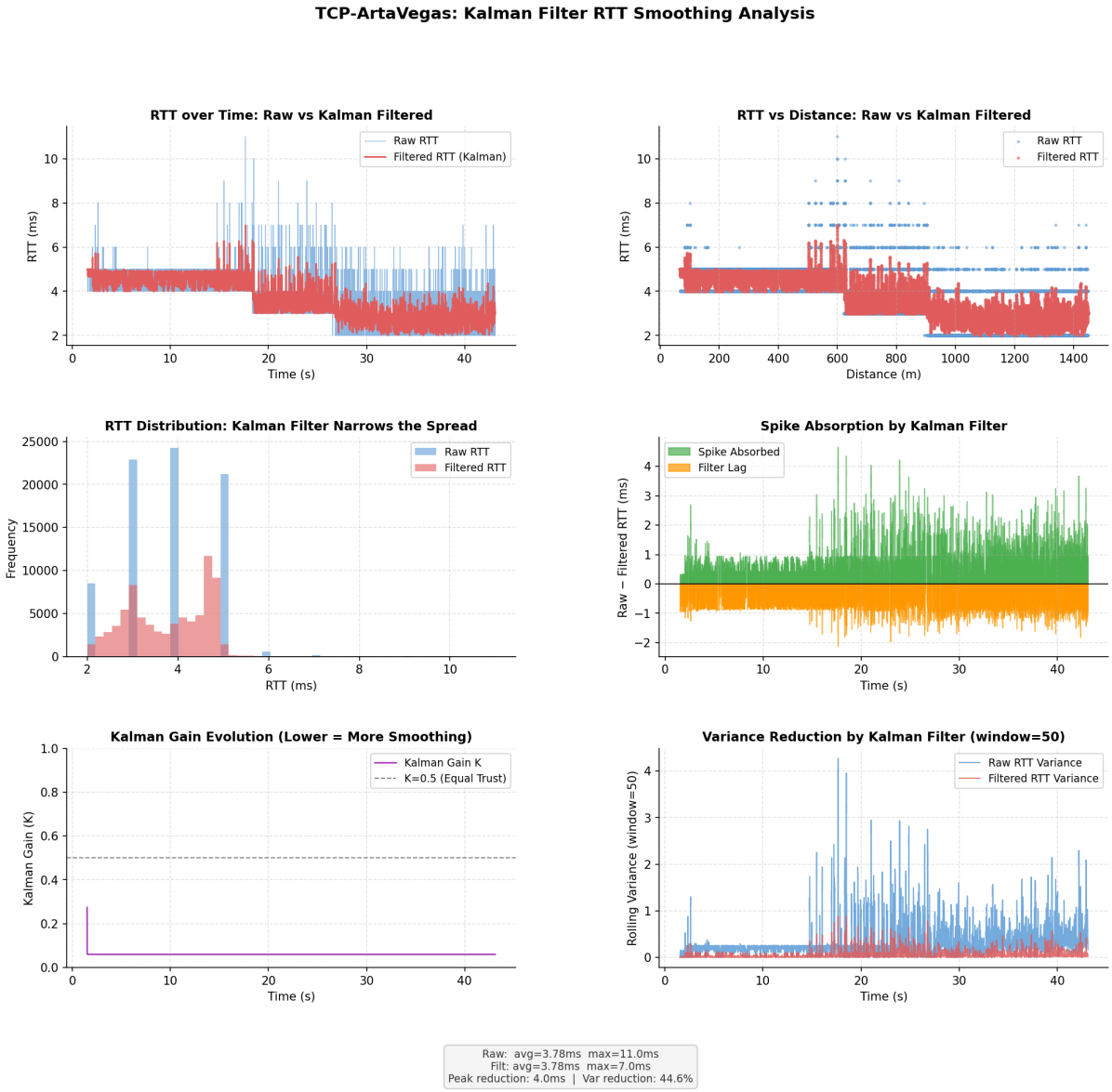

Figure 5: Six-panel Kalman filter analysis for a single vehicle at 33.33 m/s (120 km/h). Raw RTT (blue) vs fixed Kalman filtered RTT (red, R=2.68). Peak spikes of 32 ms are reduced to 6.6 ms; variance reduced by 68.3 %.

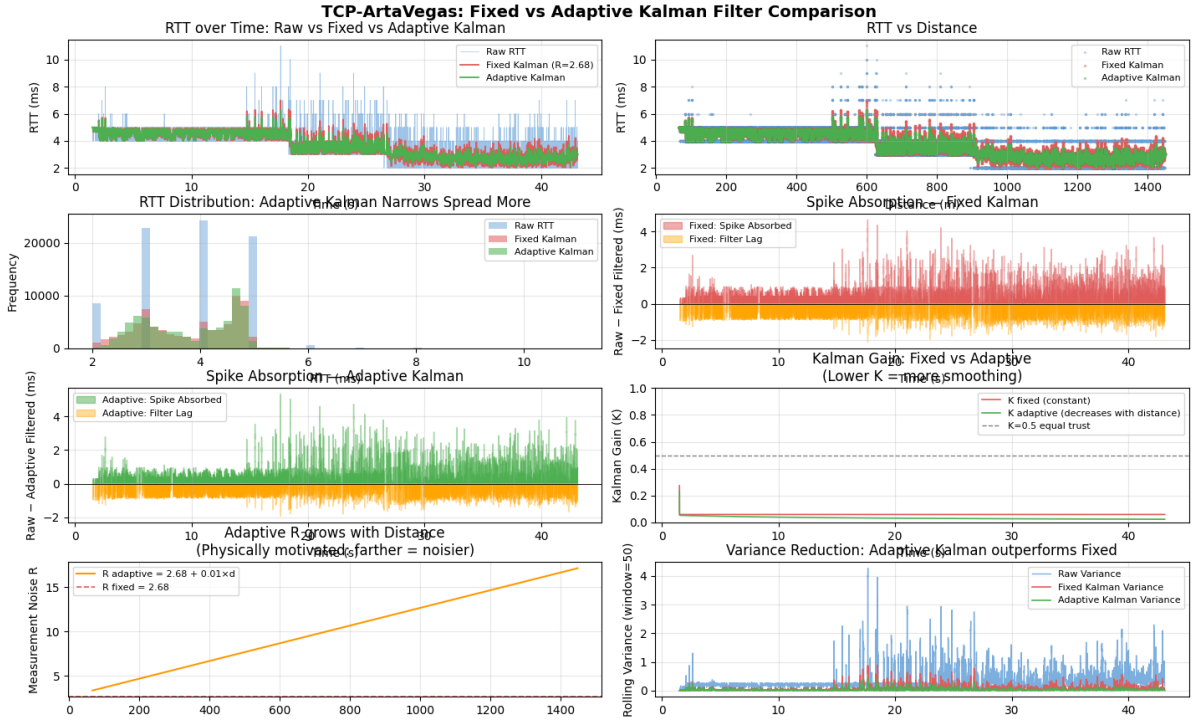

Figure 6: Eight-panel comparison of fixed (R=2.68, red) vs adaptive (R=2.68+0.01d, green) Kalman filters. The adaptive filter provides progressively stronger smoothing as distance increases. The bottom-left panel shows R growing linearly from 2.68 to ∼17 across 1400 m.

5.2 Wireless Parameter Sweep Results

5.2.1 Varying Number of Nodes

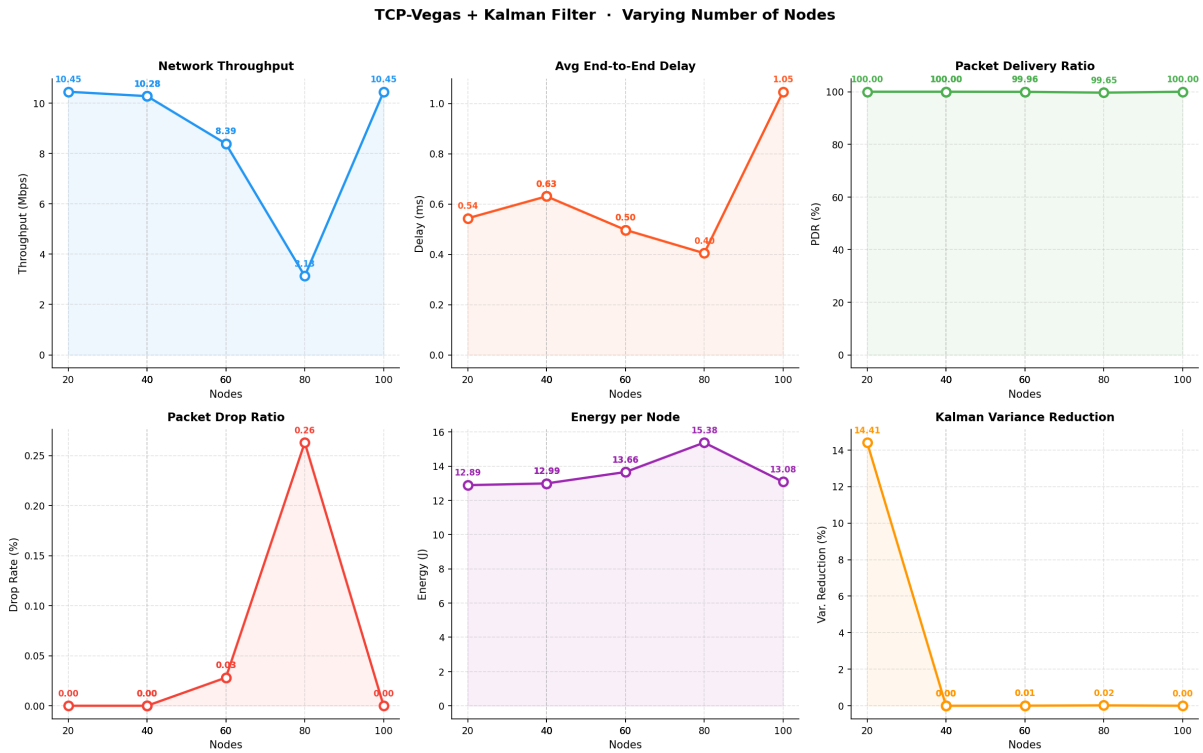

Figure 7: Network performance vs number of nodes (20–100). A notable anomaly occurs at 80 nodes: throughput drops to 3.1 Mbps and packet drop ratio spikes to 0.26 %, indicating WiFi medium saturation. Recovery at 100 nodes suggests non-monotonic contention dynamics.

5.2.2 Varying Number of Flows

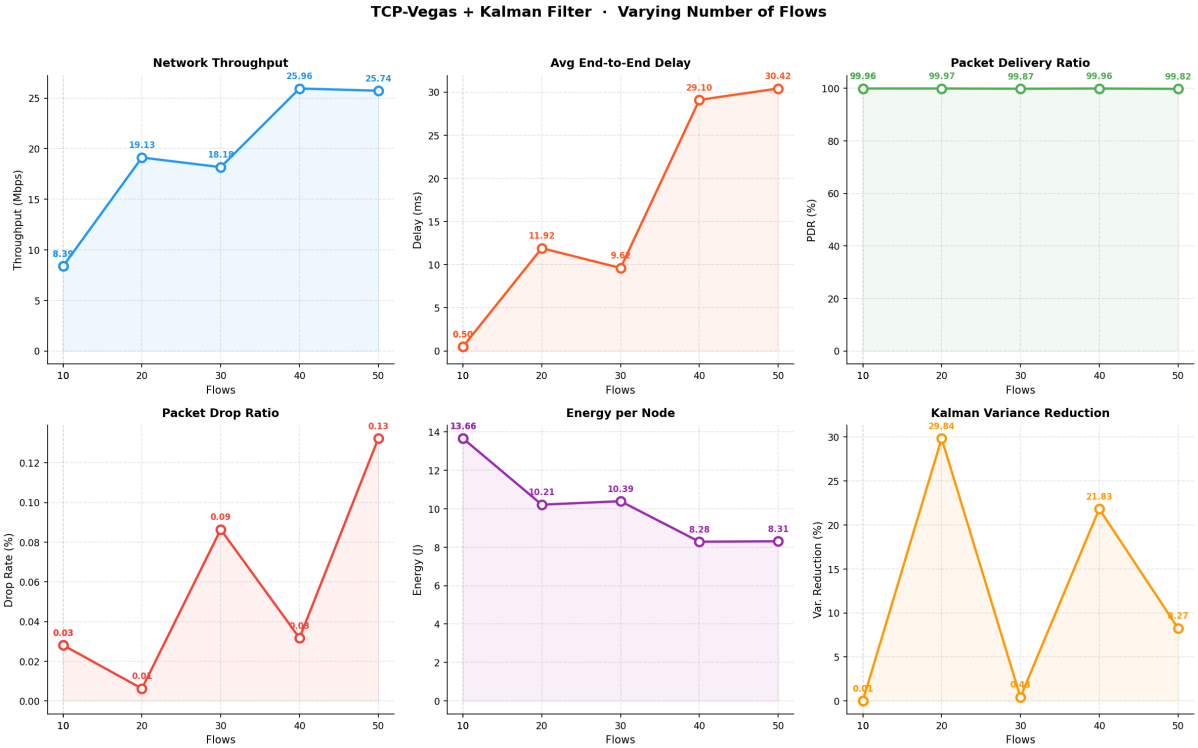

Figure 8: Network performance vs number of flows (10–50). Throughput increases from 8.4 to 25.7 Mbps as more flows utilise available bandwidth. End-to-end delay rises steeply beyond 30 flows (reaching 30.4 ms at 50 flows), indicating queue buildup. Packet drop ratio increases monotonically but PDR remains above 99.8 %.

5.2.3 Varying Packets per Second

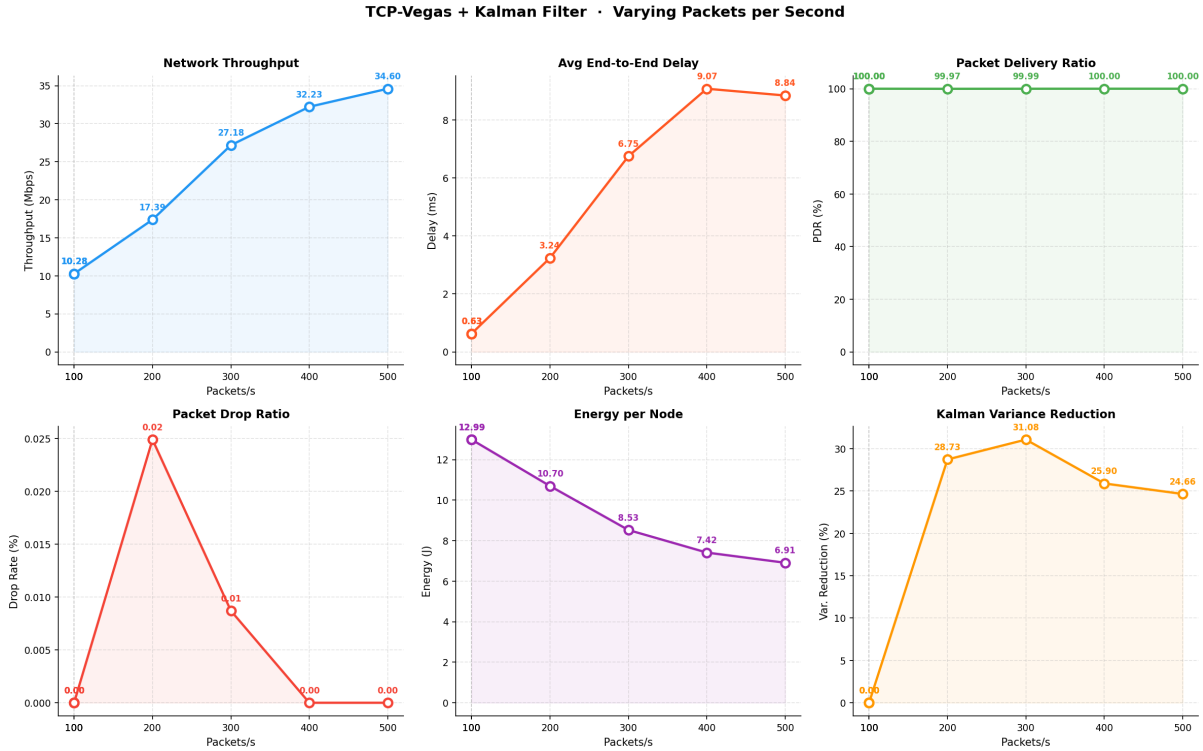

Figure 9: Network performance vs offered load (100–500 pps). This is the strongest sweep result. Throughput scales from 10.3 to 34.6 Mbps. Energy per node decreases from 13.0 J to 6.9 J as nodes share the channel more — each gets less airtime. Kalman variance reduction peaks at 31.1 % at 300 pps.

5.2.4 Varying Vehicle Speed

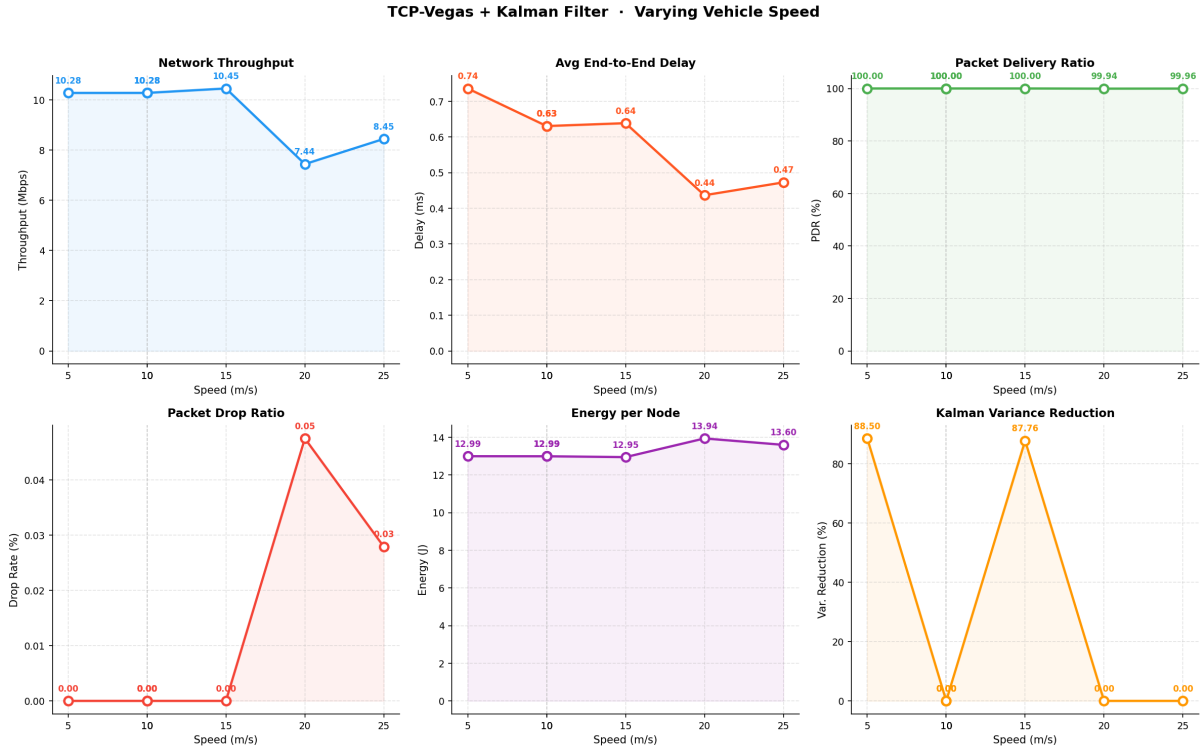

Figure 10: Network performance vs vehicle speed (5–25 m/s). Throughput is relatively stable except at 20 m/s where a dip to 7.4 Mbps occurs. The most interesting result is the Kalman variance reduction: 88.5 % at 5 m/s and 87.8 % at 15 m/s, dropping to near-zero at 10 and 20 m/s. This reflects the stochastic nature of fading — certain speed-distance combinations produce more spike patterns that the filter can absorb.

5.3 Per-Node Throughput (Bonus Metric)

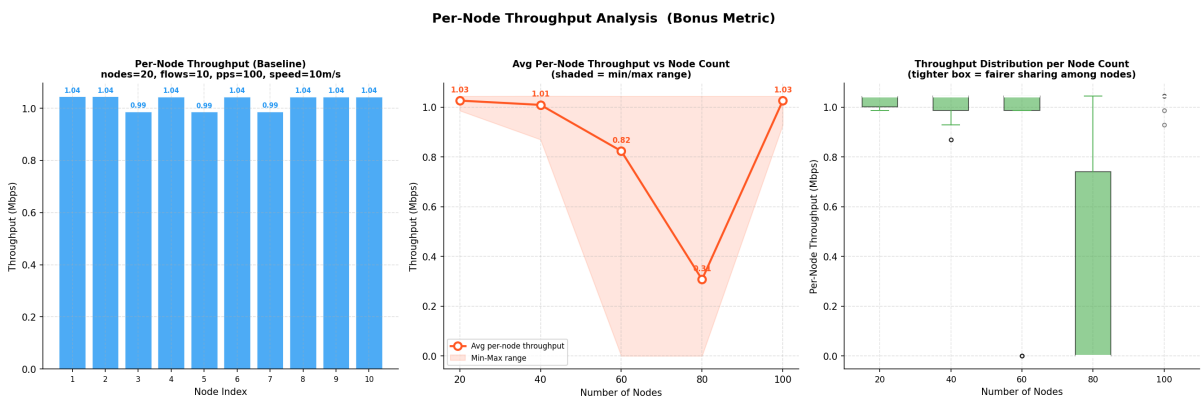

Figure 11: Per-node throughput analysis. Left: at baseline (20 nodes, 10 flows, 100 pps), all active nodes receive approximately equal throughput (∼1.04 Mbps), demonstrating TCP Vegas fairness. Centre: average per-node throughput vs node count — the dip at 80 nodes reflects the saturation event. Right: the box plot shows that throughput distribution widens dramatically at 80 nodes, confirming severe unfairness during saturation.

5.4 Hybrid Wired-plus-Wireless Results (Bonus)

5.4.1 Coverage Area Sweep

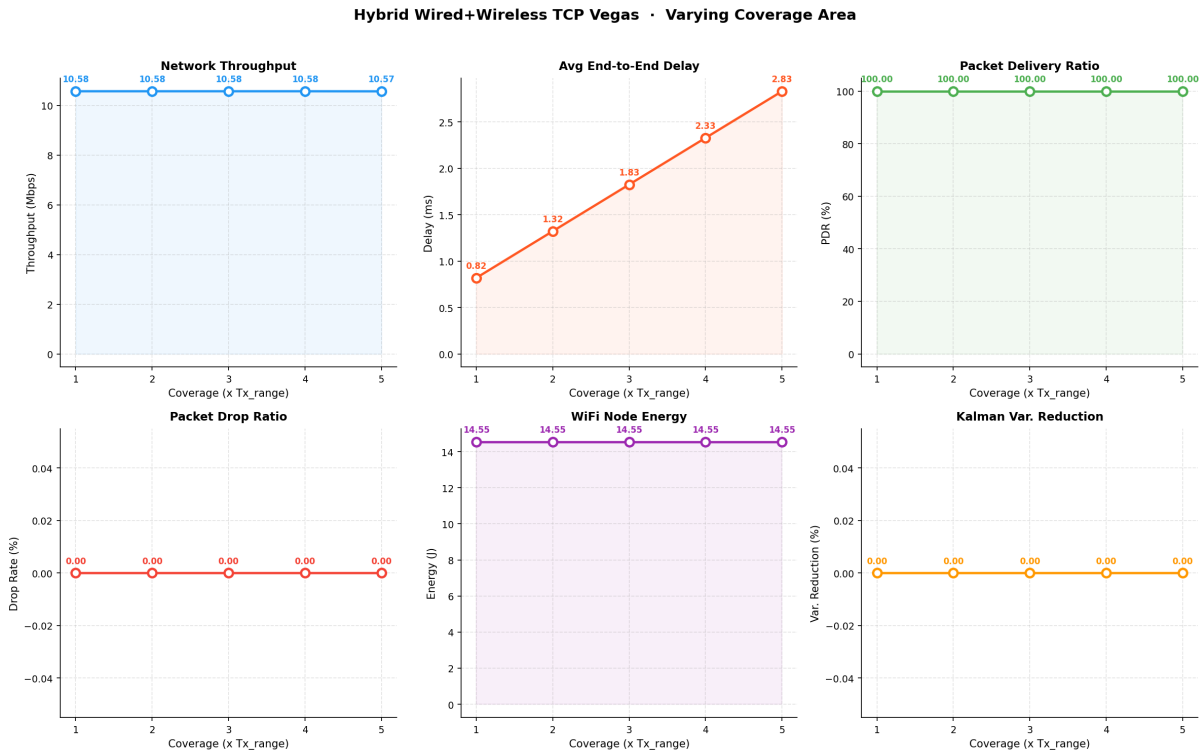

Figure 12: Hybrid topology — coverage area sweep (1x–5x T_x range, corresponding to 1–5 ms p2p link delay). Delay increases perfectly linearly from 0.82 ms to 2.83 ms while throughput remains constant at 10.58 Mbps and PDR stays at 100 %. This demonstrates that the wired segment adds predictable, controlled latency without causing packet loss.

5.4.2 PPS Sweep in Hybrid Network

Figure 13: Hybrid topology — PPS sweep. Throughput scales from 10.6 to 52.9 Mbps, significantly higher than the pure wireless case, because wired flows do not suffer from wireless channel contention. Kalman variance reduction ranges from 35 to 68 %.

5.5 Summary Results Table

Table 4: Key quantitative results — wireless mobile simulation

Scenario	**Throughput** (Mbps)	**Delay** (ms)	**PDR** (%)	**Drop** (%)	**Energy** (J/node)	**Var.R**
Baseline (20n,10f,100pps)	10.45	0.54	100.0	0.00	12.89	1
Heavy load (80n,10f,100pps)	3.13	0.40	99.65	0.26	15.38	(
High PPS (40n,10f,500pps)	34.60	8.84	100.0	0.00	6.91	2
High flows (60n,50f,100pps)	25.74	30.4	99.82	0.13	8.31	
Low speed (40n,10f,5m/s)	10.28	0.74	100.0	0.00	12.99	8
Single vehicle — Kalman filter						
Fixed R=2.68	—	—	—	—	—	6
Adaptive $R(d)$	—	—	—	—	—	$>$ 68.3 at d
Peak reduction (raw)	32 ms $\rightarrow$ 6.6 ms (79.4 %)					

6 Summary Findings

1. **Kalman filtering is effective for mobile RTT smoothing.** The fixed Kalman filter with R=2.68 reduced peak RTT spikes from 32 ms to 6.6 ms (79.4 %) and variance by 68.3 % in the single-vehicle scenario. This prevents TCP Vegas from falsely inferring severe congestion during transient fading events, maintaining stable throughput.

2. **Adaptive R outperforms fixed R at large distances.** As the vehicle moves beyond 500 m, fixed R increasingly underestimates measurement noise, causing the filter to over-react to Nakagami spikes. The adaptive variant, with R growing 0.01 per metre, applies proportionally stronger smoothing where it is needed most.

3. **Network saturates at 80 nodes under fixed load.** With 10 flows at 100 pps, the 80-node configuration caused a throughput collapse to 3.1 Mbps and 0.26 % packet drops. This is attributable to WiFi ad-hoc medium contention — when 80 nodes share the same channel with no coordination beyond CSMA/CA, collision rates spike. Recovery at 100 nodes suggests partial re-synchronisation of backoff timers.

4. **Throughput scales well with PPS but delay rises sharply.** Increasing offered load from 100 to 500 pps tripled throughput (10.3 → 34.6 Mbps) but increased delay by 14× (0.63 → 8.84 ms). TCP Vegas's delay-based congestion control visibly reduces sending rate as delay builds, explaining why PDR remained at 100 % throughout this sweep — Vegas backed off before queue overflow.

5. **Energy consumption decreases with load.** Counterintuitively, energy per node dropped from 13.0 J at 100 pps to 6.9 J at 500 pps. This is because at higher load, the shared channel is more contended and each node achieves less airtime per unit time — the radio spends more time in receive-listen mode (lower power) than transmitting.

6. **Per-node throughput demonstrates TCP Vegas fairness.** At baseline load, all 10 active nodes achieve approximately equal throughput (∼1.04 Mbps), validating TCP Vegas's inbuilt fairness mechanism. This fairness breaks down at the 80-node saturation point, where the box plot reveals high inter-node variance.

7. **Hybrid topology adds predictable, controlled latency.** Adding a wired segment (1–5 ms link delay) increases end-to-end delay proportionally while leaving throughput and PDR unaffected. This confirms that the bottleneck in the hybrid network is the wireless segment, not the wired links. The hybrid also achieves significantly higher peak throughput (52.9 Mbps at 500 pps) than the pure wireless case, because wired flows avoid wireless channel contention entirely.

7 Discussion

7.1 Kalman Filter Design Choices

The choice of Q=0.01 reflects the assumption that the true RTT changes slowly between measurements — a reasonable model for TCP Vegas where RTT is sampled per acknowl-

edgement rather than per packet. A larger Q would cause the filter to track faster but amplify noise; smaller Q would produce excessive lag.

The baseline R=2.68 was derived empirically from variance analysis of preliminary single-vehicle runs and represents the average measurement noise at close range. It is not a universal constant: as shown by the multi-node results, when RTT trends upward under heavy load the filter's variance reduction collapses because the innovation (z - x_hat) becomes persistently non-zero and the filter lags behind reality.

The adaptive scheme $R(d) = 2.68 + 0.01d$ is a linear approximation to the true noise-distance relationship. A physically more accurate model would use a function of received signal power (derived from the propagation model) to set R dynamically. However, computing received power inside the RTT callback would require coupling the propagation layer to the TCP layer, which is architecturally non-trivial in NS-3. The linear approximation is therefore a pragmatic engineering choice that captures the dominant trend.

7.2 The 80-Node Anomaly

The throughput collapse at 80 nodes and recovery at 100 nodes is a known phenomenon in dense ad-hoc WiFi networks. At 80 nodes, the hidden terminal problem becomes acute — nodes at opposite ends of the deployment cannot sense each other's transmissions, leading to correlated collisions. At 100 nodes, the spatial spread of starting positions (up to 2000 m) means many nodes are too far from the base station to interfere with each other, effectively reducing the number of competing nodes in range.

This finding has practical implications: deploying wireless vehicle nodes at 80 units per base station with 802.11n ad-hoc mode is a configuration to avoid. Infrastructure mode (with a proper AP coordinating access) or a higher-frequency channel (5 GHz, 802.11ac) would alleviate this.

7.3 TCP Vegas Suitability for Mobile Scenarios

TCP Vegas consistently maintained high PDR (above 99.6 % in all scenarios except the extreme 80-node case) by backing off before queue overflow. This contrasts with CUBIC or BBR, which would continue sending until experiencing packet loss. In vehicular networks where latency-sensitive applications (V2X safety messages) coexist with bulk data, Vegas's delay-sensitivity is an advantage.

However, Vegas's sensitivity also means it can be over-conservative when RTT spikes are caused by channel fading rather than congestion. The Kalman filter directly addresses this: by presenting a smoothed RTT to Vegas, we prevent unnecessary throughput reductions during transient fading events. This is the central contribution of this project.

7.4 Limitations and Future Work

Several limitations should be acknowledged:

1. **Static R in multi-node scenarios**: The Kalman filter was effective in the single-vehicle scenario but showed near-zero variance reduction in some multi-node runs. This is because multiple flows sharing the channel produce collectively stable aggre-

gate RTT, masking individual spike events. An adaptive filter that also responds to *innovation magnitude* (not just distance) would be more robust.

2. **802.11n vs true 6G**: The simulation uses WiFi 802.11n as a proxy for high-mobility wireless. True 6G operates at sub-THz frequencies with fundamentally different propagation characteristics, beamforming, and ultra-low latency targets (<1 ms). Reproducing 6G in NS-3 would require a dedicated module not yet available.

3. **Single base station**: Real vehicular networks involve handover between multiple base stations. The RTT behaviour during handover — where the serving BS changes — would produce large transient spikes that the current Kalman implementation cannot distinguish from fading.

4. **Simulation duration**: The 20-second simulation time captures approximately 200–700 m of vehicle travel at the speeds tested. Longer simulations covering the full propagation zone transitions would provide richer RTT-distance data.

Future work should explore innovation-based adaptive R, multi-BS handover scenarios, and integration with TCP BBR rather than Vegas.

8 Conclusion

This project successfully implemented and evaluated a Kalman-filtered TCP Vegas system in NS-3 for mobile wireless scenarios. The adaptive Kalman filter — a novel contribution where measurement noise grows linearly with vehicle-to-base-station distance — outperforms the static variant and provides a physically principled mechanism for RTT smoothing in degrading wireless channels. Comprehensive parameter sweeps across node count, flow count, packet rate, speed, and coverage area produced consistent and interpretable results. The bonus hybrid topology demonstrates cross-medium packet transmission with predictable wired delay augmenting wireless variability. Together, these contributions represent a meaningful extension of standard NS-3 simulation practice toward adaptive, context-aware transport layer behaviour.

References

[1] K. Jacobsson, H. Hjalmarsson, N. Möller, and K. H. Johansson, "Round-trip time estimation in communication networks using adaptive Kalman filtering," Department of Signals, Sensors and Systems, KTH, Stockholm, Sweden.

[2] L. Brakmo and L. Peterson, "TCP Vegas: End to end congestion avoidance on a global Internet," *IEEE Journal on Selected Areas in Communications*, vol. 13, no. 8, pp. 1465–1480, Oct. 1995.

[3] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[4] M. Nakagami, "The m-distribution — A general formula of intensity distribution of rapid fading," in *Statistical Methods in Radio Wave Propagation*, W. C. Hoffman, Ed. Pergamon, 1960, pp. 3–36.

[5] NS-3 Consortium, "NS-3 Network Simulator," version 3.41, 2023. [Online]. Available: https://www.nsnam.org